[Def] Let $D \in \mathbb{Z}$ be a square-free integer.

Then, a set $\mathbb{Q}(\sqrt{D}) = \{a+b\sqrt{D} \mid a,b \in \mathbb{Q}\}$ forms a field, and

$\mathbb{Z}[\sqrt{D}] = \{a+b\sqrt{D} \mid a,b \in \mathbb{Z}\}$ forms a Ring, as a subring of $\mathbb{Q}(\sqrt{D})$.

Further, if $D \equiv 1 \bmod 4$, then for $w = \dfrac{1+\sqrt{D}}{2}$,

$$\mathbb{Z}[w] = \{a+bw \mid a,b \in \mathbb{Z}\}$$

is also subring of $\mathbb{Q}(\sqrt{D})$, and contains $\mathbb{Z}[\sqrt{D}]$. Figure out how.

At first, $\mathbb{Q}(\sqrt{D})$ is vector space over $\mathbb{Q}$, with basis $\{1, \sqrt{D}\}$. So, define a Norm as

$$N : \mathbb{Q}(\sqrt{D}) \to \mathbb{Q} : a+b\sqrt{D} \mapsto a^2 - D\cdot b^2$$

Actually, $N(\alpha) = \prod_{\sigma \in Gal(\mathbb{Q}(\sqrt{D}),\mathbb{Q})} \sigma(\alpha)$. So, if $\alpha \neq 0$, $\alpha^{-1} = \dfrac{\overline{\alpha}}{N(\alpha)}$.

In $\mathbb{Q}(\sqrt{D})$, It is clearly abelian group under the addition, and by the fact above, $\mathbb{Q}(\sqrt{D}) - \{0\}$ is abelian group under the multiplication.

Now, let's to dissect $\mathbb{Z}[\sqrt{D}]$. Clearly $\mathbb{Z}[\sqrt{D}]$ is closed under the substraction.

And, for $a+b\sqrt{D}$, $c+d\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$,

$(a+b\sqrt{D})\cdot(c+d\sqrt{D}) = ac + bd\cdot D + (ad+bc)\cdot\sqrt{D}$, so closed under the multiplication.

Associative, Commutative, Containing 0 and 1 are clear.

Moreover, for $\mathbb{Z}[w]$, it is clearly group under the addition, and,

$$\left(a + b\cdot\frac{1+\sqrt{D}}{2}\right)\cdot\left(c + d\cdot\frac{1+\sqrt{D}}{2}\right) = ac + bd\cdot\frac{D-1}{4} + (ad+bc+bd)\cdot\frac{1+\sqrt{D}}{2}$$

is contained in $\mathbb{Z}[w]$, since $(D-1)/4 \in \mathbb{Z}$, by $D \equiv 1 \bmod 4$.

(More observation: for $a+b\sqrt{D}$ with $b \neq 0$, a minimal polynomial over the given Ring is

$$(x-a)^2 - D\cdot b^2 = x^2 - 2ax + a^2 - D\cdot b^2$$

which has another root as $a-b\sqrt{D}$.)

[Thm. The field Norm $N: \mathbb{Q}(\sqrt{D}) \mapsto \mathbb{Q}; a+b\sqrt{D} \mapsto a^2 - Db^2$ preserves the multiplication.

Proof. Let $a+b\sqrt{D}, c+d\sqrt{D} \in \mathbb{Q}(\sqrt{D})$ be given. Then,

$$\begin{aligned} N([a+b\sqrt{D}]\cdot[c+d\sqrt{D}]) &= N(ac+bdD+(ad+bc)\sqrt{D}) \\ &= (ac+bdD)^2 - D\cdot(ad+bc)^2 \\ &= a^2c^2+b^2d^2D^2+2abcd\cdot D - D\cdot(a^2d^2+b^2c^2+2abcd) \\ &= a^2c^2+b^2d^2D^2 - D\cdot(a^2d^2+b^2c^2) \\ &= (a^2-Db^2)\cdot(c^2-Dd^2) \\ &= N(a+b\sqrt{D})\cdot N(c+d\sqrt{D}). \end{aligned}$$

[Thm. In $\mathbb{Z}[\sqrt{D}]$ or $\mathbb{Z}[\omega]$, an element α is a unit iff $N(\alpha) = \pm 1$.

Proof. If α is unit, then $N(\alpha\cdot\alpha^{-1}) = N(\alpha)\cdot N(\alpha^{-1}) = 1$.

But, since $N(\alpha), N(\alpha^{-1}) \in \mathbb{Z}$, both are 1 or both are -1.

Conversely, if $N(\alpha) = \pm 1$, $\frac{\overline{\alpha}}{N(\alpha)}$ is inverse of α.